AE 153 – Agricultural Meteorology & Climatology (2,1,2)

Lecture 1

E. Bessah (PhD)
Department of Agricultural and Biosystems Engineering
College of Engineering
KNUST, Kumasi

Course Content

- Importance and practical applications of agro-meteorology measurements
- Climatological data processing and analysis
- Climate variability and change

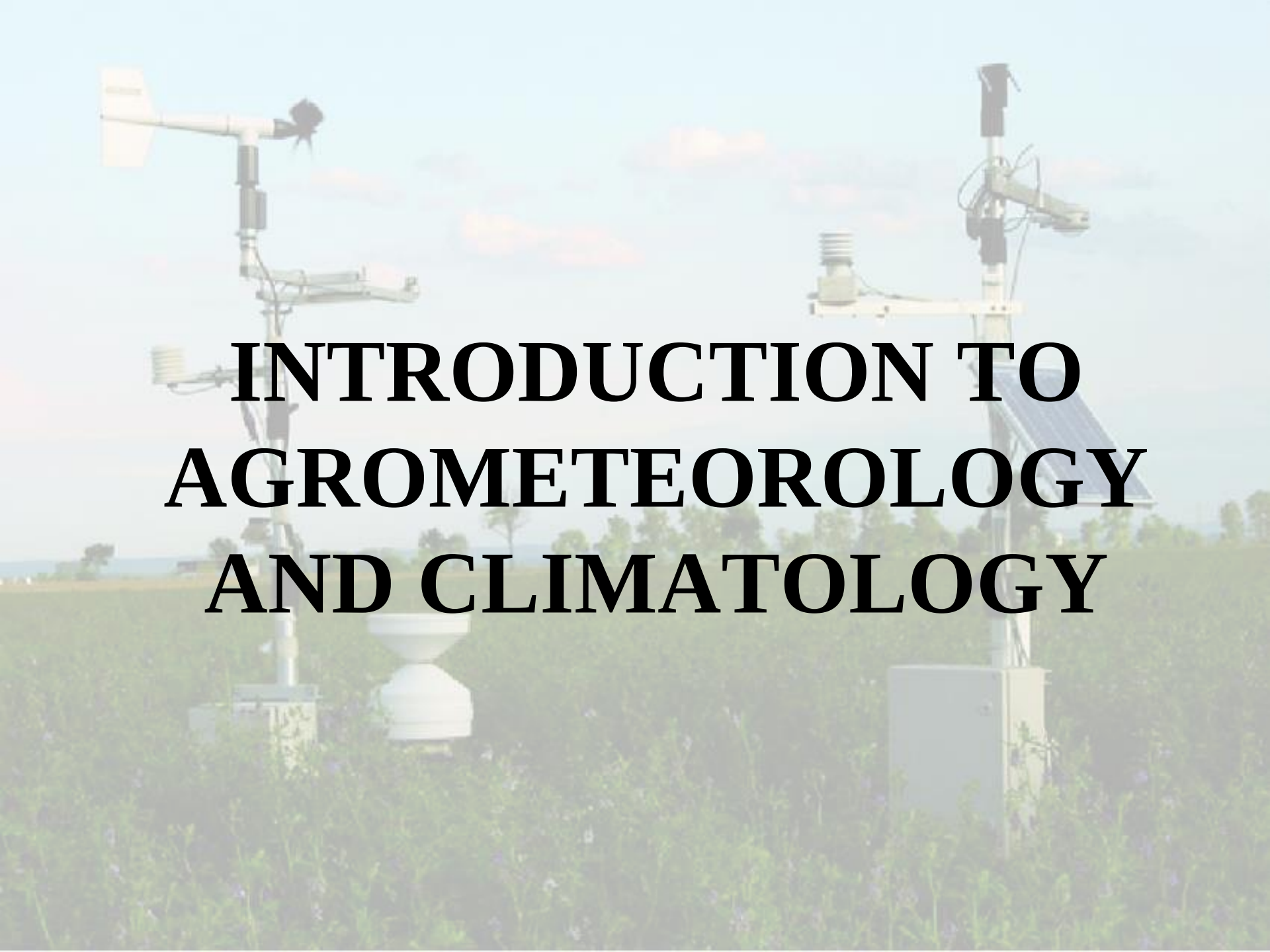
Course Objectives

Specific Objectives:

1. To measure agro-meteorology data
2. To determine the practical applications of the agro-meteorology data
3. To process and analyze weather/climate data
4. To explain solar radiation relationship with the atmosphere
5. To explain climate variability and climate change

Reference Material

1. Soil and water conservation engineering by G. O. Schwab
2. Hydrology in practice by E. M. Shaw
3. Agrometeorology: principles and applications of climate studies in agriculture by H S Mavi and G. J. Tupper



INTRODUCTION TO AGROMETEOROLOGY AND CLIMATOLOGY

Teminologies

Weather: The **state of the atmosphere at a given time and place**, with respect to variables such as **rainfall, temperature, moisture, wind velocity, and atmospheric pressure**.

That is the present condition of these variables or elements and their variations over periods of up to say two weeks.

Climate: Is the **statistics of variables** (such as temperature, humidity, atmospheric pressure, wind, rainfall and other meteorological elements in a given region over long period of time (usually 30 years).

Factors that determine/influences the climate of a location:

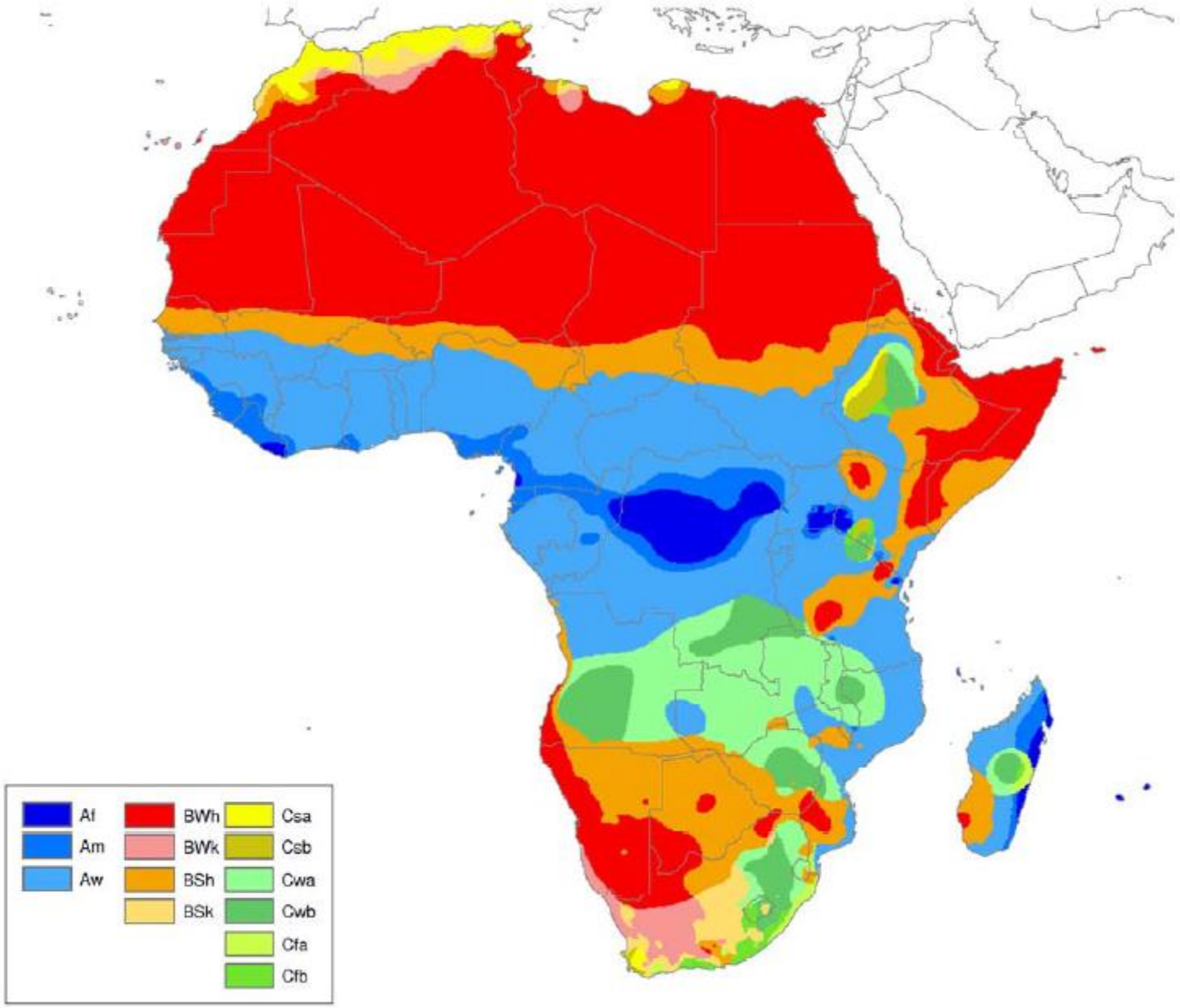
- Latitude
- Terrain
- Vegetation
- Altitude
- Presence of water bodies and their currents

Climate classification

Is done using the average climatic conditions and the typical ranges of different variables, most commonly using temperature and precipitation.

The most commonly used classification scheme was originally developed by Köppen and Thornthwaite.

- Köppen classification: The five main groups are: A (Tropical); B (Dry); C (Temperate); D (Continental) and E (Polar)



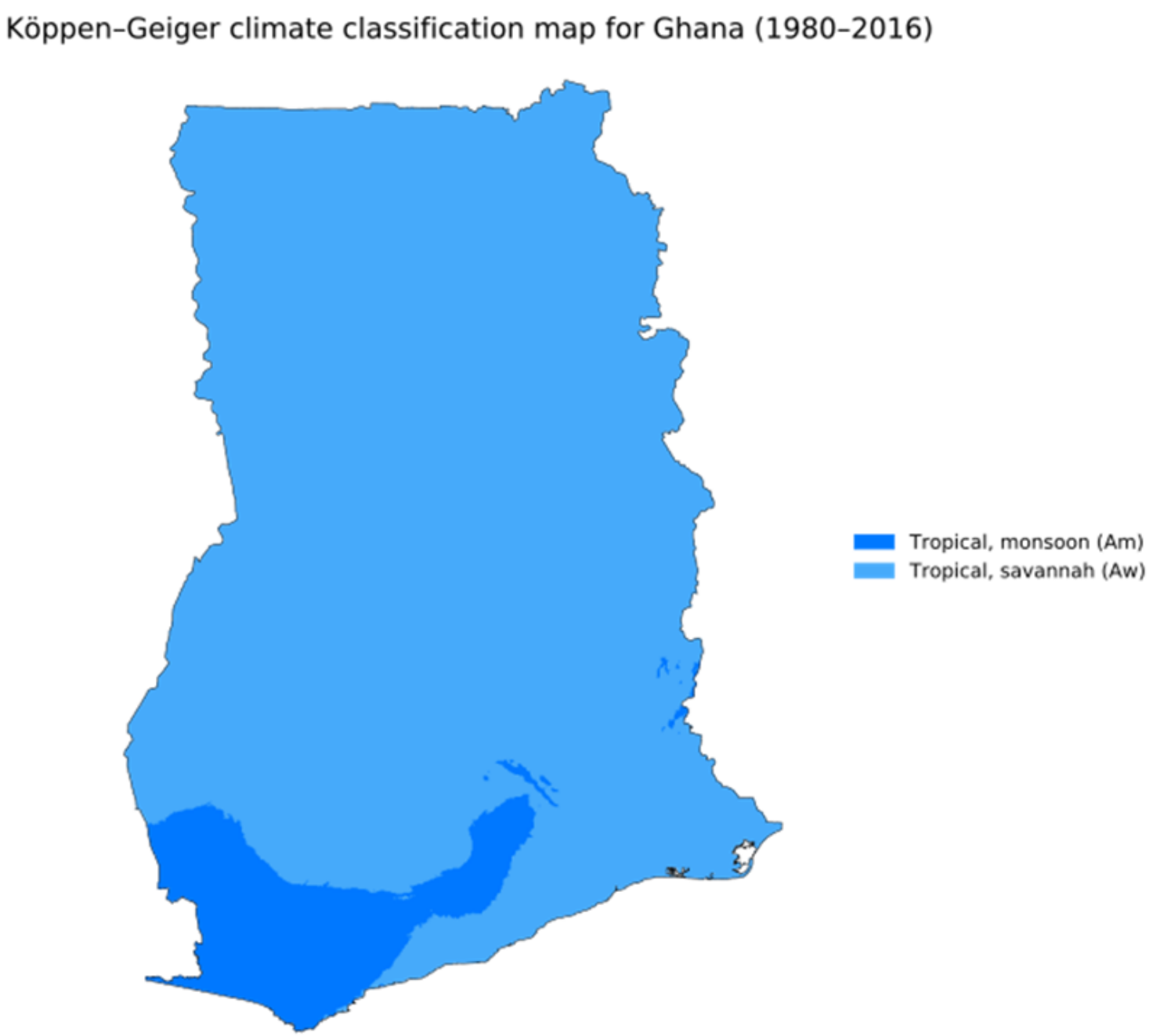
Source: Peel et al., 2007

Table 1. Description of Köppen climate symbols and defining criteria.

1st	2nd	3rd	Description	Criteria*
A			Tropical	$T_{cold} \geq 18$
	f		- Rainforest	$P_{dry} \geq 60$
	m		- Monsoon	Not (Af) & $P_{dry} \geq 100 - MAP/25$
	w		- Savannah	Not (Af) & $P_{dry} < 100 - MAP/25$
B			Arid	$MAP < 10 \times P_{threshold}$
	W		- Desert	$MAP < 5 \times P_{threshold}$
	S		- Steppe	$MAP \geq 5 \times P_{threshold}$
		h	- Hot	$MAT \geq 18$
		k	- Cold	$MAT < 18$
C			Temperate	$T_{hot} > 10$ & $0 < T_{cold} < 18$
	s		- Dry Summer	$P_{sdry} < 40$ & $P_{sdry} < P_{wwet}/3$
	w		- Dry Winter	$P_{wdry} < P_{swet}/10$
	f		- Without dry season	Not (Cs) or (Cw)
		a	- Hot Summer	$T_{hot} \geq 22$
		b	- Warm Summer	Not (a) & $T_{mon10} \geq 4$
		c	- Cold Summer	Not (a or b) & $1 \leq T_{mon10} < 4$
D			Cold	$T_{hot} > 10$ & $T_{cold} \leq 0$
	s		- Dry Summer	$P_{sdry} < 40$ & $P_{sdry} < P_{wwet}/3$
	w		- Dry Winter	$P_{wdry} < P_{swet}/10$
	f		- Without dry season	Not (Ds) or (Dw)
		a	- Hot Summer	$T_{hot} \geq 22$
		b	- Warm Summer	Not (a) & $T_{mon10} \geq 4$
		c	- Cold Summer	Not (a, b or d)
		d	- Very Cold Winter	Not (a or b) & $T_{cold} < -38$
E			Polar	$T_{hot} < 10$
	T		- Tundra	$T_{hot} > 0$
	F		- Frost	$T_{hot} \leq 0$

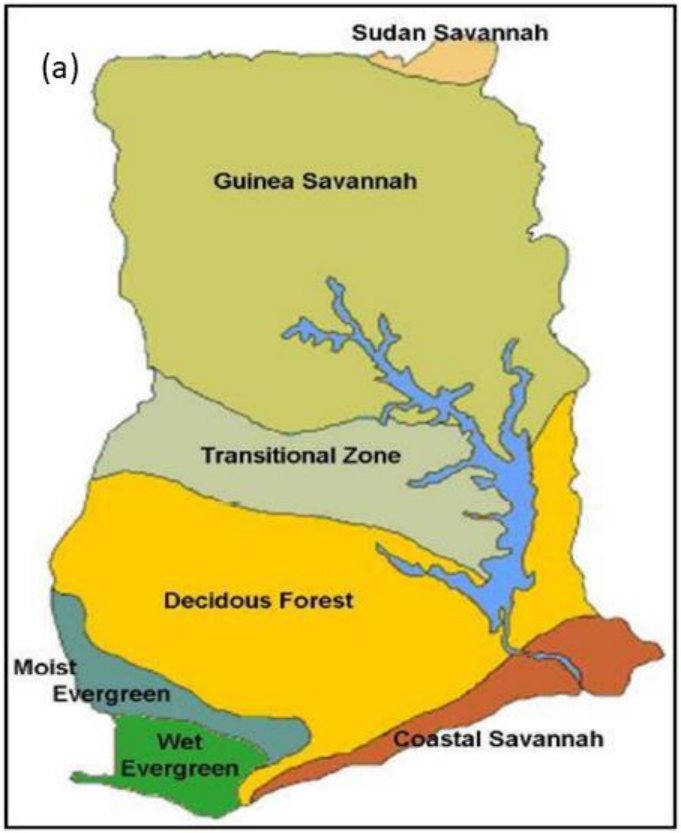
*MAP = mean annual precipitation, MAT = mean annual temperature, T_{hot} = temperature of the hottest month, T_{cold} = temperature of the coldest month, T_{mon10} = number of months where the temperature is above 10, P_{dry} = precipitation of the driest month, P_{sdry} = precipitation of the driest month in summer, P_{wdry} = precipitation of the driest month in winter, P_{swet} = precipitation of the wettest month in summer, P_{wwet} = precipitation of the wettest month in winter, $P_{threshold}$ = varies according to the following rules (if 70% of MAP occurs in winter then $P_{threshold}$ = 2 x MAT, if 70% of MAP occurs in summer then $P_{threshold}$ = 2 x MAT + 28, otherwise $P_{threshold}$ = 2 x MAT + 14). Summer (winter) is defined as the warmer (cooler) six month period of ONDJFM and AMJJAS.

Source: Peel et al., 2007

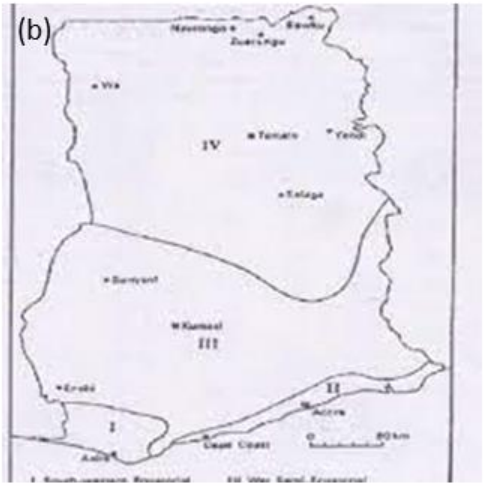


(Source: Beck et al., 2018)

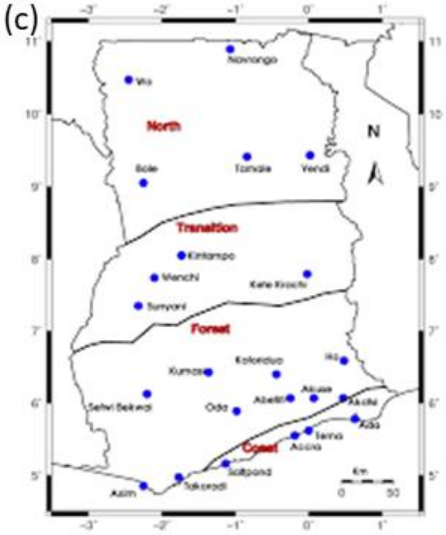
Existing agro-ecological and climatic zones in Ghana



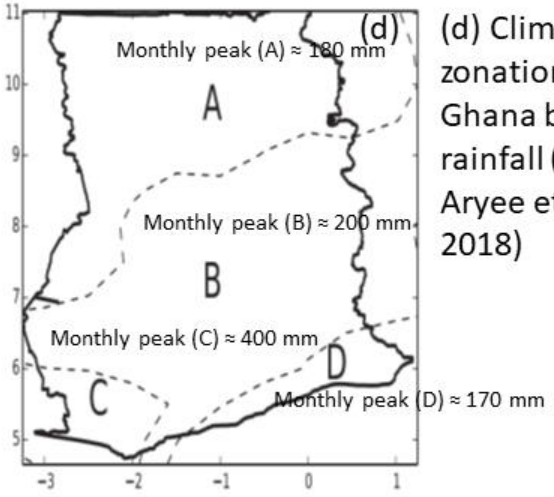
(a) FAO agro-ecological zones of Ghana (Source: Asravor et al., 2019).



(b) Climatic Regions of Ghana (Source: Abass, 2009)
I - South-western Equatorial
II - Dry Equatorial
III - Wet semi-Equatorial
IV - Tropical Continental or Savannah

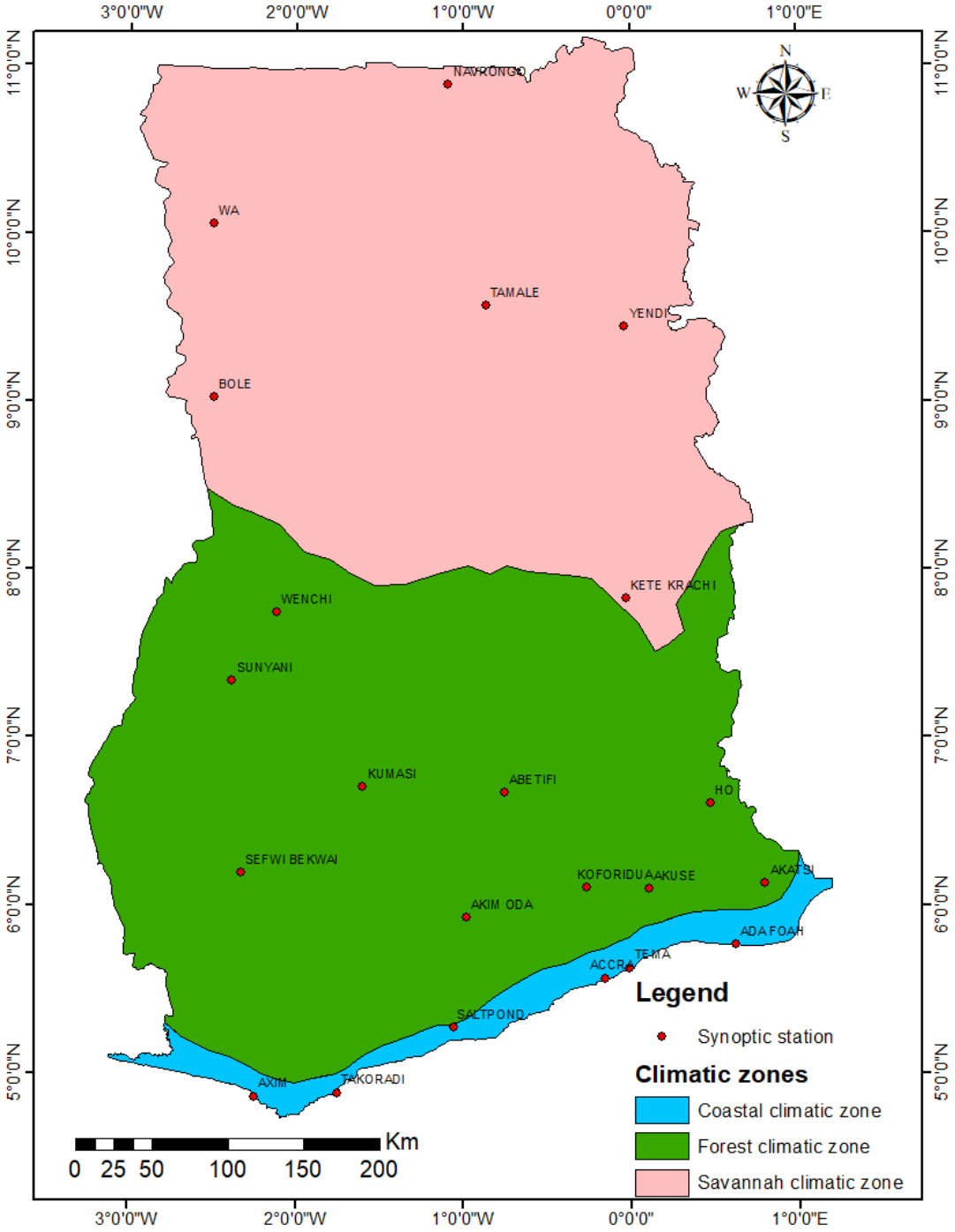


(c) Map of the Gmet stations covering the four agro-ecological zones of Ghana (Source: Amekudzi et al., 2015).



(d) Climatic zonation of Ghana based on rainfall (Source: Aryee et al., 2018)

Source: Bessah et al., 2022



Source: Bessah et al., 2022

Teminologies

Meteorology: The **scientific study of** the **atmosphere** and **atmospheric conditions**, especially as they relate to weather and weather forecasting.

Agro-meteorology: The branch of meteorology that studies the **effect of weather and climate on agriculture**. It examines the effects and impacts of weather and climate on crops, rangeland, livestock, and various agricultural operations.

Agricultural meteorologists: **collect, analyse and interpret weather and climate data** needed to understand the interactions between vegetation and animals and their atmospheric environments.

Benefits of agro-meteorological information

Agro-meteorologists:

1. provide information valuable in making proper decisions
   a. managing resources consumed by agriculture
   b. optimizing agricultural production
   c. adopting farming practices that minimizes adverse effects of agriculture on the environment
2. provide information ensuring economic and environmental sustainability of agric. now and in the future
3. quantify, evaluate and provide information on the impact and consequences of climate variability and change on agriculture
4. assist policy makers in developing strategies to deal with extreme climatic events such as floods, rain storms, or droughts and climatic changes such as global warming and climate variability

Assignment 1

Describe the five components of climate and explain why climatic parameters are used in defining climate variability and change.

Submission: 25 March 2022.